

Retail Insights Assistant

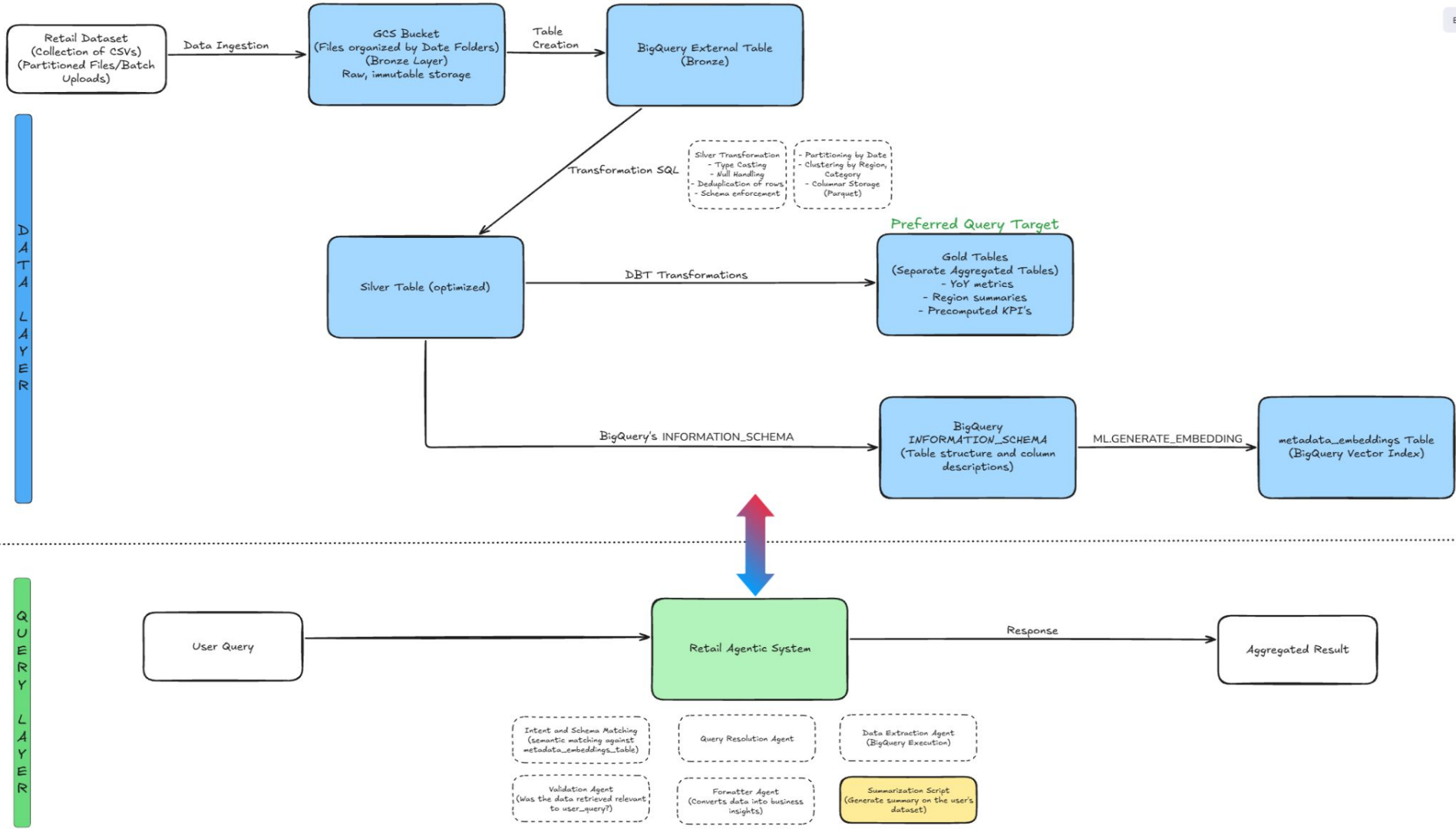
GenAI + Scalable Data System

By: Fardeen Alam



System Design for 100GB+ Data Scale





Core Tool Selections for the Data Layer



Google Cloud Storage (GCS) - Scalable Data Lake

- **Role:** Acts as the **Bronze (Raw) Layer**; stores immutable 100GB+ datasets at a fraction of the cost of database storage.
- **Key Choice:** Enables **Decoupled Ingestion**; BigQuery can query these files directly via **External Tables** without the need for complex, manual loading scripts.

BigQuery - Serverless Data Warehouse

- **Role:** Powers our **Silver (Cleaned)** and **Gold (Aggregated)** layers.
- **Key Choice:** **Serverless Scaling** handles massive joins without cluster management. Native **BigQuery ML Vector Search** enables the "Retail Assistant" to perform semantic retrieval directly where the data lives.

dbt (Data Build Tool) - SQL Transformation

- **Role:** Orchestrates the transformation from **Silver -> Gold**.
- **Key Choice:** Brings **Software Engineering** to SQL (Version control, testing, and modularity). Ensures the data feeding the LLM is high-quality, deduplicated, and business-validated.



Architecture Techniques: Data Layer Optimization

Medallion Architecture (Bronze -> Silver -> Gold)

- **Strategy:** Multi-stage data refinement.
- **Benefit:** **Bronze** preserves raw data integrity; **Silver** provides a cleaned "Single Source of Truth"; **Gold** stores high-level aggregates for instant, low-cost summary generation.

Table Partitioning (Time-Based Sharding)

- **Strategy:** Physically dividing the 100GB table by **Order_Date**.
- **Benefit:** When the Assistant queries a specific month, BigQuery ignores 90%+ of the dataset, **slashing query costs and latency** significantly.

Table Clustering (Multi-Column Sorting)

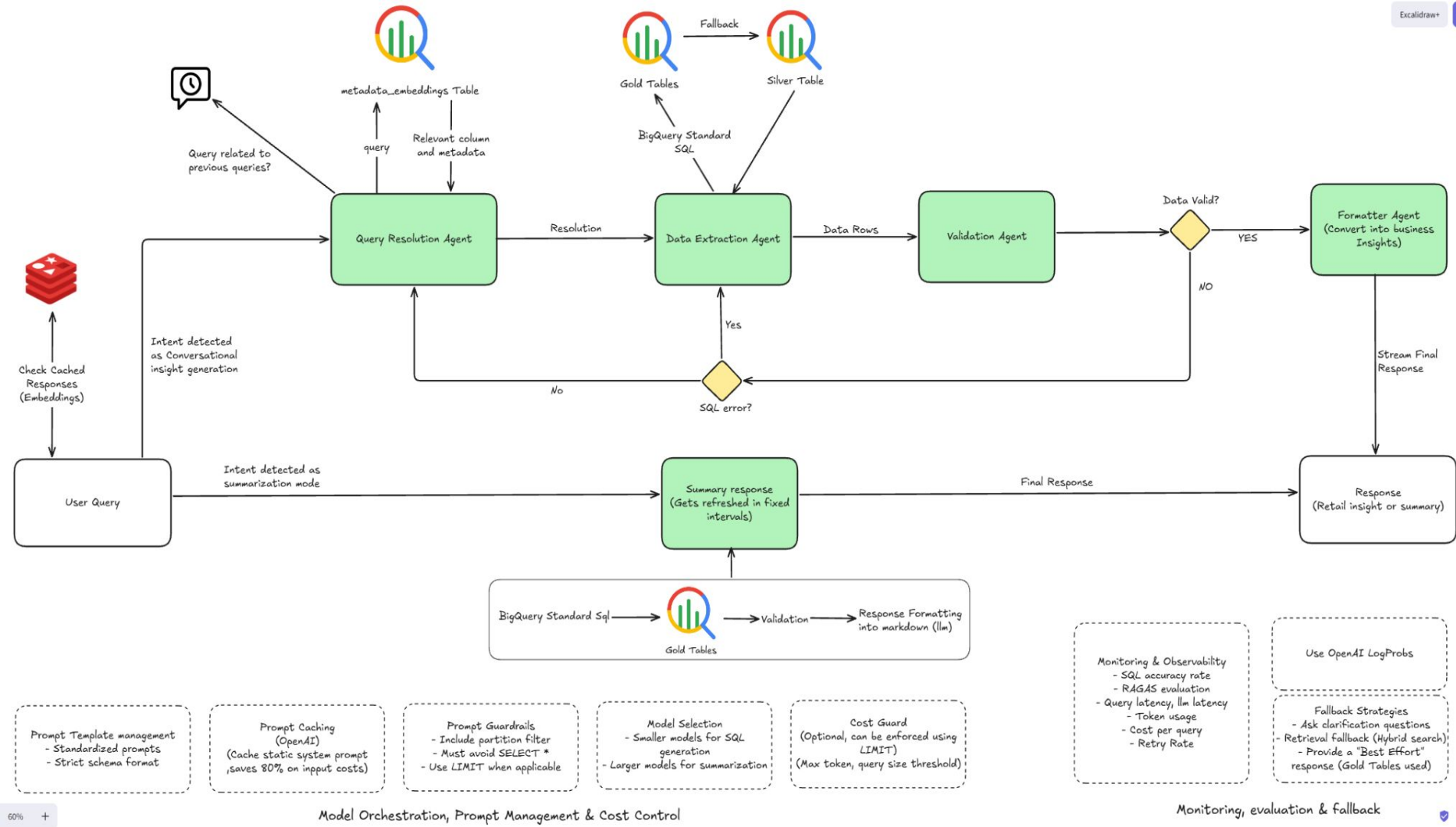
- **Strategy:** Sorting data within partitions by high-frequency filters like **Region** or **Category**.
- **Benefit:** Physically co-locates related data, allowing for **lightning-fast "drill-down" queries** (e.g., "Sales for Electronics in the North").

BigQuery Vector Indexing

- **Strategy:** Creating an embedding-based index on table metadata and product/column descriptions.
- **Benefit:** Enables **Semantic Retrieval**; the agent can find relevant columns or products based on "meaning" rather than just exact keyword matches.

Retail Agentic System Architecture (100GB+)





Agentic Layer: Scaling & Orchestration



LangGraph Orchestration (Stateful Multi-Agent Workflow)

- **Technique:** Uses a cyclic, state-based graph to manage agent handovers.
- **Scale Benefit:** Prevents "hallucination loops" by using a **Validation Agent** as a circuit breaker to catch and fix incorrect SQL before execution.

Semantic Context Caching

- **Technique:** Bolds and caches the massive 100GB schema metadata in the LLM's context window.
- **Scale Benefit:** Reduces latency and **cuts token costs by up to 80%** by reusing schema definitions across multiple user queries.

Metadata-Driven Schema Linking

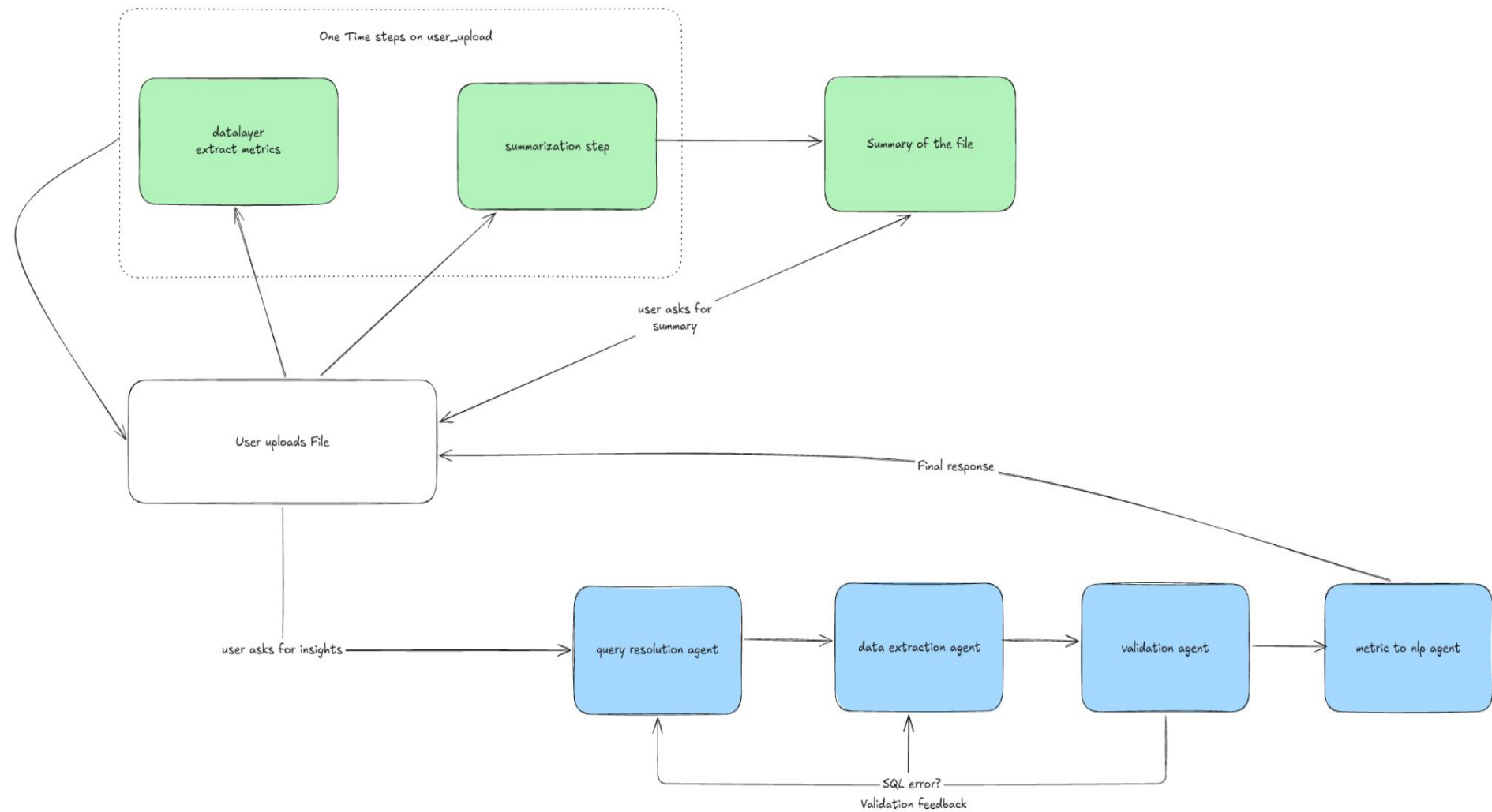
- **Technique:** Agents perform a **Vector Search** on table metadata (not the raw data) to identify relevant columns.
- **Scale Benefit:** Allows the LLM to navigate thousands of partitions and hundreds of columns without exceeding context limits or scanning irrelevant data.

Automated Partition Injection

- **Technique:** The **Extraction Agent** is hard-coded to require date/region filters derived from the user query.
- **Scale Benefit:** Ensures every generated SQL query is "Scale-Aware," preventing expensive and slow **Full Table Scans** on the 100GB dataset.

System Architecture and data flow for MVP (Current System)





Query-Response Pipeline



Data Ingestion & Automated Summarization



Data Layer (DuckDB & GCS Integration):

- **Action:** User uploads CSV via **Streamlit**; data is loaded into an in-memory **DuckDB** table for high-performance processing.
- **Profiling:** Runs a full schema scan (dtypes, null counts, min/max/avg) to produce a **TableProfile**.

Metadata Grounding:

- **Action:** Serializes the profile into a **Structured Metadata Context** (cardinality, sample values, ranges).
- **Benefit:** Injects this context into every agent prompt, ensuring the LLM is "grounded" in the actual dataset.

Summarization Pipeline (Parallel Processing):

- **Query Planning:** LLM generates **8-12 diverse SQL queries** (totals, trends, top-N) based on metadata.
- **Synthesis:** Executes queries in DuckDB and synthesizes results into a **Markdown Executive Report** (Executive Summary, Key Metrics, Trends).

Conversational Multi-Agent Q&A Workflow



Query Resolution Agent (The Planner):

- Converts user questions into a structured **Query Spec** (intent, filters, aggregations).
- Maintains **Conversation Memory** (last 5 turns) to handle follow-up questions efficiently.

Data Extraction Agent (The Coder):

- Translates the spec into precise **DuckDB SQL** and executes it.
- Automatically handles upto 200MB scale through optimized SQL generation.

Validation Agent (The Quality Gate):

- Audits the SQL and results against the user's intent.
- **Self-Correction:** Triggers **up to 3 independent retries** by routing back to either the Planner (for logic errors) or Coder (for SQL errors).

Formatter Agent (The Analyst):

- Transforms raw table rows into **concise, business-friendly natural language** answers.

Evidence





Limit 200MB per file • CSV

Browse files



Amazon Sale Report.csv 65.5MB



TABLE

upload

ROWS

128,975

COLUMNS

24

NUMERIC COLS

0



Type *summary* to see the full auto-generated data summary, or ask any analytical question about your data.

what are the cities with great potential?

RETAIL INSIGHTS

Bengaluru stands out as the city with the greatest potential, generating approximately ₹6.85 million in revenue. Following Bengaluru, Hyderabad shows strong promise with around ₹4.95 million. Mumbai, New Delhi, and Chennai also contribute significantly, with revenues of ₹3.70 million, ₹3.61 million, and ₹3.10 million respectively. These cities demonstrate a robust market presence, indicating strong opportunities for growth and investment.

Ask a question about your data ...

Send

summarize orders for delhi

RETAIL INSIGHTS

In Delhi, there were a total of 336 distinct orders placed, resulting in 324 items sold. The total revenue generated from these orders amounted to approximately INR 216,986. This indicates a strong level of activity and sales in the region.

what are the cities with great potential?

RETAIL INSIGHTS

Bengaluru shows the greatest potential with the highest total revenue of approximately INR 6.85 million, along with over 10,000 orders. Following closely are Hyderabad and Mumbai, with total revenues of around INR 4.95 million and INR 3.70 million, respectively. New Delhi and Chennai also demonstrate strong performance, contributing significantly to overall sales. These cities are key markets to focus on for future growth opportunities.

In which area should I expand business?

RETAIL INSIGHTS

You should consider expanding your business in Bengaluru, which has the highest total sales amount of approximately INR 6.85 million, along with a strong order count of over 10,400. Following Bengaluru, Hyderabad also shows significant potential with around INR 4.95 million in sales and over 7,400 orders. Other notable cities include Mumbai and New Delhi, but their sales figures are considerably lower. Overall, focusing on Bengaluru and Hyderabad could yield the best results for expansion.

RETAIL INSIGHTS

Executive Summary This dataset provides insights into sales performance, order fulfillment, and customer behavior across various categories and regions. It encompasses a total of 120,378 orders with a revenue of approximately INR 78.59 million. The data covers transactions up to the month of June 2022.

Key Metrics

- **Total Orders:** 120,378
- **Total Transactions:** 128,975
- **Total Quantity Sold:** 116,649
- **Total Revenue:** INR 78,592,678
- **Average Order Value:** INR 652.88

Top Performers

Top 5 Categories by Revenue

Category	Revenue (INR)	Quantity Sold	Orders
Set	39,204,124	45,289	47,845
Kurta	21,299,547	45,045	46,561
Western Dress	11,216,073	13,943	14,994
Top	5,247,592	9,992	10,155

Retail Insights

AI-POWERED



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Amazon Sale Report.csv 65.5MB



FILE

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Thank You

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